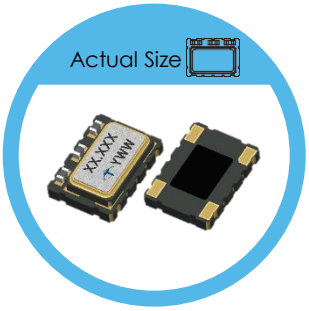
TAITIEN
Our Performance, Your Reputation

TT-F Type Ultra Wide Temperature
7.0 x 5.0 mm SMD Voltage Controlled Temperature Compensated Crystal Oscillator



RoHS Compliant

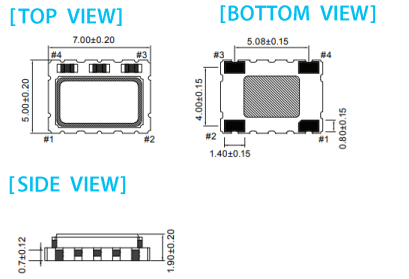
FEATURES

- Typical 7.0 x 5.0 x 1.9 mm ceramic SMD package.
- Ultra wide temp. range: -55°C to +125°C with ±0.2 to 0.5 ppm stability
- CMOS and Clipped Sine wave (without DC-cut capacitor) output optional.

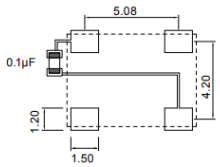
TYPICAL APPLICATION

- Femtocell , Base Stations
- WLAN/WiMAX/WIFI, Wireless Communications

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



Recommended Soldering Pattern
To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between V_{DD} and GND Pads.

PIN FUNCTION (mm)

PIN#	FUNCTION
1	V_{CON}:VC-TCXO Floating:TCXO
2	GND
3	Output
4	V_{DD}

ELECTRICAL SPECIFICATION

Parameter			3.3V Min.	3.3V Typ.	3.3V Max.	Unit	Test Condition
Supply Voltage Variation (V_{DD})			V_{DD} – 5%	V_{DD}	V_{DD} + 5%	V	
Frequency Range			10	-	52	MHz	
Standard Frequency			10, 14.4, 20, 28.8, 38.88			MHz	Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position.
Frequency Tolerance			-	-	±1.5	ppm	Frequency at 25°C, 1 hour after reflow.
Frequency Stability	vs Supply Voltage (±5%) Change		-	-	±0.1	ppm	
	vs Load (±10%) Change		-	-	±0.1	ppm	
	vs Aging (@1ST Year)		-	-	±1.0	ppm/year	
Supply Current	CMOS		-	-	8.0	mA	
	Clipped Sine Wave		-	-	5.0	mA	
Output Level	CMOS	Output High (Logic "1")	90%V_{DD}	-	-	V	
		Output Low (Logic "0")	-	-	10%V_{DD}	V	
		Duty	45	-	55	%	
	Clipped Sine Wave		1.2	-	-	V_{p-p}	
Load	CMOS		-	15pF	-		
	Clipped Sine Wave		-	10 KΩ // 10pF	-		
Control Voltage Range (VCTCXO)			0.3	-	3.0	V	
Pulling Range (VCTCXO)			-13	-	+6.5	ppm	
Pulling Linearity			-	-	12	%	
Vc Input Impedance (VCTCXO)			100	-	-	KΩ	
Phase Noise @ 20MHz		100 Hz	-	-122	-	dBc/Hz	
		1K Hz	-	-139	-	dBc/Hz	
		10K Hz	-	-146	-	dBc/Hz	
Start Time			-	-	10	mSec	
Storage Temp. Range			-55	-	+125	°C	

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±0.2	±0.28	±0.5
-55 ~ +95	△	○	○
-55 ~ +105	△	○	○
-55 ~ +125	X	X	○

○: Available △:Conditional X: Not available

Note: not all combination of options are available. Other specifications may be available upon request.
Specifications subject to change without notice.

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